ASSAY CATALOGUE

Mextropic
Assay Capabilities

102 CAPABILITIES / 19 CATEGORIES

Contents

Imaging and omics datasets **3**
- Imaging 3
- Clinical Imaging 3
- Genomics / Transcriptomics 5
- Epigenomics 5
- Proteomics 6
- Metabolomics 7

Wet-lab validation **8**
- Genetic Engineering 8
- Biochemical Screening 8
- Binding & Interaction 9
- Cell-Based Assays 9
- Pharmacology 10
- Safety Pharmacology & Genotoxicity 10
- ADMET 10
- Antibody / Protein Developability 11
- Protein Engineering 12
- Protein Expression 12
- Chemistry 12
- Microbiology 12
- Synthetic Biology 12

Glossary **13**

Index of assays **17**

Imaging and omics datasets

Imaging

High-content imaging
High-content imaging with multiplexed, image-based feature extraction on high-content screening platforms

Spatial transcriptomics
Spatially resolved transcript profiling of tissue sections, retaining the tissue coordinates of gene expression

Spatial proteomics (imaging MS)
Mass-spectrometry imaging of tissue sections, producing spatially resolved protein and peptide distribution maps

Histopathology and tissue staining
Histological preparation, microscopic assessment, and selected IHC/ICC workflows

Clinical Imaging

Availability, delivery route, and the ethics and consent basis for human imaging data are confirmed per programme.

MRI - structural
T1-, T2- and FLAIR-weighted magnetic resonance series covering soft-tissue and organ anatomy, delivered as DICOM

MRI - diffusion-weighted and DTI
Diffusion-weighted imaging and diffusion-tensor acquisitions with derived ADC and tractography maps

MRI - functional (BOLD)
Blood-oxygen-level-dependent functional MRI, task-based and resting-state

MRI - spectroscopy (MRS)
In vivo magnetic resonance spectroscopy for regional metabolite concentrations

Cardiac MRI
Cine, perfusion and late-gadolinium-enhancement cardiac magnetic resonance for function, viability and fibrosis

CT - non-contrast and contrast-enhanced
Multi-detector computed tomography series, with and without iodinated contrast

CT angiography
ECG-synchronised contrast-enhanced CT of coronary and peripheral vasculature

Cone-beam CT
Cone-beam computed tomography acquired in interventional and image-guided settings

Digital radiography (X-ray)
Planar digital projection radiography across chest, skeletal and abdominal indications

Fluoroscopy and interventional angiography
Real-time fluoroscopic and catheter-angiographic series from diagnostic and interventional procedures

Mammography and digital breast tomosynthesis
Full-field digital mammography and tomosynthesis volumes

Dual-energy X-ray absorptiometry (DEXA)
Bone mineral density and body-composition scans

Ultrasound - B-mode and Doppler
Greyscale and colour/spectral Doppler ultrasound across abdominal, vascular and obstetric indications

Echocardiography
Transthoracic two-dimensional, three-dimensional and strain echocardiography

Ultrasound elastography
Shear-wave and strain elastography for tissue stiffness, including hepatic fibrosis staging

PET/CT
Positron-emission tomography with co-registered CT, including FDG and non-FDG tracers

PET/MRI
Simultaneous positron-emission tomography and magnetic resonance acquisition

SPECT and SPECT/CT
Single-photon-emission computed tomography, with co-registered CT where acquired

Retinal imaging - fundus and OCT
Colour fundus photography and optical coherence tomography of the retina

Digital pathology / whole-slide imaging
Scanned whole-slide H&E and immunohistochemistry images for quantitative review

Genomics / Transcriptomics

Bulk RNA-seq
Bulk RNA sequencing on current-generation Illumina short-read platforms

Single-cell / single-nucleus RNA-seq
Single-cell and single-nucleus RNA sequencing on Chromium-based platforms

Long-read / full-length isoform sequencing
Long-read and full-length isoform sequencing on PacBio HiFi (Revio, Sequel II) and Oxford Nanopore platforms

Small RNA-seq (miRNA)
Small RNA and miRNA sequencing as a standard NGS application

Targeted RNA panels
Targeted RNA panel sequencing on NGS platforms

Epigenomics

ChIP-seq
ChIP-seq for genome-wide protein–DNA binding site mapping

CUT&RUN / CUT&Tag
CUT&RUN and CUT&Tag chromatin profiling using established kit chemistries

ATAC-seq
ATAC-seq for genome-wide chromatin accessibility profiling

Bisulfite sequencing / WGBS
WGBS and RRBS methylation sequencing for base-resolution DNA methylation mapping

Hi-C / chromosome conformation capture
Proximity-ligation sequencing to map genome-wide chromatin contacts, for 3D genome architecture and for scaffolding genome assemblies

Proteomics

Bottom-up (shotgun) proteomics

LC-MS/MS shotgun proteomics on Orbitrap Exploris and Q Exactive-class instruments for protein identification and quantification

Targeted proteomics (MRM/SRM/PRM)

Targeted MRM, SRM and PRM proteomics for quantitative protein panels

Top-down / middle-down proteomics

Intact-protein and middle-down mass spectrometry for proteoform-level characterisation, resolving PTM combinations on the intact molecule rather than inferring them from peptides

PTM-focused proteomics (phospho/glyco/ubiquitin/acetyl)

PTM-enrichment proteomics covering phospho, glyco, ubiquitin and acetyl modifications

Single-cell proteomics

Nanoscale sample preparation coupled to high-sensitivity LC-MS/MS, quantifying proteomes from individual cells

Proximity labeling (BioID/APEX)

Enzyme-catalysed proximity biotinylation from a bait protein, followed by streptavidin enrichment and LC-MS/MS, to map the proximal interactome in living cells

Crosslinking-MS (XL-MS)

Chemical crosslinking followed by LC-MS/MS identification of crosslinked peptide pairs, giving residue-level distance restraints on protein complex architecture

HDX-MS

Hydrogen–deuterium exchange with LC-MS readout, mapping backbone solvent accessibility for conformational analysis and epitope-level characterisation

Chemical proteomics / ABPP

Activity-based probe labelling with LC-MS/MS readout, reporting enzyme activity and target engagement across the native proteome, including competition formats for engagement confirmation

Metaproteomics

LC-MS/MS profiling of microbial community samples, with taxonomic and functional assignment of the identified proteins

Metabolomics

LC-MS based metabolomics (untargeted/targeted)

Untargeted and targeted LC-MS metabolomics with feature detection and quantification

GC-MS based metabolomics

GC-MS metabolomics for volatile and derivatised metabolite profiling

NMR-based metabolomics

NMR-based metabolomics on high-field instrumentation (600 MHz and above)

Fluxomics

Stable-isotope (13C) labelling with LC-MS/GC-MS readout and metabolic flux modelling, resolving pathway-level flux distributions

Wet-lab validation

Genetic Engineering

CRISPR knock-out/knock-in gene editing
CRISPR knock-out and knock-in editing at 50-85% editing efficiency, with sequence-verified clones in 3-5 weeks and in-house off-target panels

Genome-wide / boutique CRISPR & RNAi screens
Pooled CRISPR and RNAi screening with custom and boutique library designs

Knockout/knock-in phenotypic characterisation
Replicate-level phenotypic characterisation of engineered lines, with automated imaging readouts

Biochemical Screening

High-throughput biochemical screening (HTS)
Automated 384-well high-throughput screening at throughputs of tens of thousands of wells per day, with compound library access through partner networks

Enzyme inhibition assays (IC50/Ki)
Enzyme inhibition profiling reporting IC50 and Ki with dose-response and fit-quality metrics

Kinase activity assays
Kinase activity and inhibition assays run against in-house panels of tens to ~100 kinases

Protease activity assays
Fluorogenic and colorimetric protease activity assays

ATPase / GTPase activity assays
ATPase and GTPase activity assays in standard biochemical formats

Fragment-based screening (NMR)
Ligand-observed NMR screening (STD, WaterLOGSY) of curated fragment libraries, detecting weak binders for fragment-based lead discovery

Binding & Interaction

SPR / BLI / MST (binding kinetics)
Binding kinetics and affinity determination by SPR and BLI on Biacore and Octet instrumentation

ITC (isothermal titration calorimetry)
Isothermal titration calorimetry for label-free binding affinity and thermodynamics

TR-FRET assays
TR-FRET homogeneous proximity assays in kit-based formats with plate-reader detection

AlphaScreen / AlphaLISA proximity assays
AlphaScreen and AlphaLISA proximity assays using standardised kits and plate-reader detection

Protein-protein interaction mapping
Protein–protein interaction mapping by targeted co-IP and pulldown

Thermal shift / DSF for hit-binding confirmation
Thermal shift and nanoDSF hit-binding confirmation on Prometheus and real-time PCR-based instrumentation

Cell-Based Assays

Phenotypic cell-based screening
Phenotypic cell-based screening at hundreds to low thousands of compounds per week

Cell-based functional / receptor assays
FLIPR-class kinetic functional assays for cell-based receptor pharmacology

Reporter gene assays
Reporter gene assay panels for transcriptional and pathway readouts

Cell viability / cytotoxicity assays
Kit-based cell viability and cytotoxicity assays

Apoptosis assays
Flow cytometry-based apoptosis assays

Flow cytometry and immunophenotyping
Cell-population and marker-expression analysis with defined panels and gating strategies

Pharmacology

Dose-response / EC50 / IC50 determination
Dose-response curve fitting with EC50 and IC50 determination and replicate-level reporting

Functional agonist/antagonist assays
Functional agonist and antagonist pharmacology assays

Off-target selectivity panels
Off-target selectivity panels, configured per programme

In vivo PK/PD, xenograft efficacy studies
In vivo PK/PD and xenograft efficacy studies in regulated animal-facility settings

Tumour and tissue pharmacodynamics
Tissue-based target-engagement and response assays aligned with pharmacology studies

Quantitative bioanalysis
Small- and large-molecule measurement in blood, plasma, urine, tissue, and other selected biological matrices

Safety Pharmacology & Genotoxicity

Electrophysiology / patch clamp (hERG, ion channels)
Automated patch-clamp electrophysiology for hERG and ion-channel liability assessment on multi-well parallel-recording platforms

Genotoxicity screening (Ames test)
Ames bacterial reverse-mutation genotoxicity screening, run to OECD test guidelines

ADMET

ADMET - solubility
Kinetic and thermodynamic solubility determination as part of core DMPK profiling

ADMET - permeability (PAMPA/Caco-2)
PAMPA and Caco-2 permeability assays for passive and cell-based permeability

ADMET - metabolic stability
Microsomal and hepatocyte metabolic stability assays

ADMET - CYP450 inhibition/induction
CYP450 inhibition and induction panels

ADMET - plasma protein binding
Plasma protein binding determination

ADMET - plasma/blood stability
Plasma and blood stability assays

ADMET - efflux transporter assays
Efflux transporter assays in transfected cell lines

Antibody / Protein Developability

Aggregation propensity (SEC-HPLC)
Aggregate and fragment quantification by SEC-HPLC

Thermal stability (Tm) - proteins/antibodies
Thermal stability (Tm) determination for proteins and antibodies by nanoDSF

Hydrophobicity (HIC-HPLC)
Surface hydrophobicity assessment by HIC-HPLC

Charge variant analysis (IEX)
Charge-variant analysis by IEX and icIEF

Polyreactivity (ELISA)
ELISA-based polyreactivity panels for antibody developability

Viscosity assessment
Viscosity and formulation rheology assessment for biologics formulations

Forced degradation / stress stability studies
ICH-aligned forced degradation and stress stability studies

Epitope binning, FcRn binding
SPR-based epitope binning and FcRn binding assessment for antibody developability

Fc effector function assays (ADCC/CDC/ADCP)
Reporter-based Fc effector function assays covering ADCC, CDC and ADCP

In vitro immunogenicity (T-cell assays)
In vitro immunogenicity assessment by T-cell assays on HLA-diverse donor PBMC banks, with MAPPs peptide-presentation workflows

Protein Engineering

Protein engineering stability assays
Comparative variant-stability screening for engineered proteins

Directed evolution / library screening
Directed evolution and display-library screening

Protein Expression

Recombinant protein expression (E. coli / CHO)
Recombinant protein expression in E. coli and CHO systems, with sequence-verified, GMP-track clone banking

Chemistry

Medicinal/peptide chemistry synthesis
Medicinal and peptide chemistry synthesis of small molecules and peptides

Microbiology

Antimicrobial susceptibility testing (culture/AST)
CLSI-aligned antimicrobial susceptibility testing by culture and AST

Organism identification (MALDI-TOF)
Organism identification by MALDI-TOF mass spectrometry

Targeted qPCR resistance-gene panels
qPCR panels for targeted resistance-gene detection on real-time PCR platforms

Synthetic Biology

Metabolic pathway engineering / product-titer assays
Metabolic pathway engineering with fermentation process development and product-titer assays at in-house bioreactor scale

Glossary

ABPP	Activity-based protein profiling — chemical-proteomics approach using reactive probes to report enzyme activity in complex lysates.
ADC	Apparent diffusion coefficient, a quantitative map derived from diffusion-weighted MRI
ADCC	Antibody-dependent cellular cytotoxicity — effector-function assay measuring immune-cell killing of antibody-coated target cells.
ADMET	Absorption, distribution, metabolism, excretion and toxicity — the drug-disposition and safety assay package.
ATAC-seq	Assay for transposase-accessible chromatin with sequencing — maps open chromatin genome-wide.
BLI	Bio-layer interferometry — label-free binding-kinetics measurement on a fibre-optic biosensor.
BOLD	Blood-oxygen-level-dependent contrast, the signal underlying functional MRI
Caco-2	Human intestinal epithelial cell line used as a cell-based permeability and efflux model.
ChIP-seq	Chromatin immunoprecipitation sequencing — maps protein–DNA binding sites genome-wide.
CRISPR	Clustered regularly interspaced short palindromic repeats — nuclease-guided genome-editing system.
CT	Computed tomography
CUT&RUN / CUT&Tag	Nuclease- or transposase-tethered chromatin profiling methods; lower cell input than ChIP-seq.
CYP450	Cytochrome P450 — the enzyme family dominating hepatic drug metabolism; inhibition and induction are core ADMET endpoints.
DEXA	Dual-energy X-ray absorptiometry
DICOM	Digital Imaging and Communications in Medicine, the standard format for clinical image data
DMPK	Drug metabolism and pharmacokinetics.

DSF	Differential scanning fluorimetry (thermal shift) — measures ligand-induced protein thermal stabilisation.
DTI	Diffusion tensor imaging
EC50 / IC50	Concentration giving half-maximal effect or half-maximal inhibition.
ECG	Electrocardiogram
ELISA	Enzyme-linked immunosorbent assay.
FcRn	Neonatal Fc receptor — binding to it governs antibody serum half-life; a developability endpoint.
FDG	Fluorodeoxyglucose, the most widely used PET tracer
FLAIR	Fluid-attenuated inversion recovery, an MRI sequence suppressing free-fluid signal
GC-MS	Gas chromatography–mass spectrometry.
HDX-MS	Hydrogen–deuterium exchange mass spectrometry — maps conformation and epitopes via backbone amide exchange.
hERG	Human ether-à-go-go-related gene potassium channel — the cardiac safety liability screened by patch clamp.
Hi-C	Chromosome-conformation capture sequencing — 3D genome contact mapping, also used to scaffold genome assemblies.
HIC-HPLC	Hydrophobic-interaction HPLC — reports surface hydrophobicity of a protein therapeutic.
HLA	Human leukocyte antigen — the MHC locus; relevant to immunogenicity and peptide-presentation assays.
HPLC	High-performance liquid chromatography.
HTS	High-throughput screening.
IEX	Ion-exchange chromatography — charge-variant analysis of protein therapeutics.
ITC	Isothermal titration calorimetry — label-free measurement of binding affinity and thermodynamics.
Ki	Inhibition constant — concentration-independent measure of inhibitor potency.

LC-MS/MS	Liquid chromatography with tandem mass spectrometry.
MALDI-TOF	Matrix-assisted laser desorption/ionisation time-of-flight mass spectrometry.
MAPPs	MHC-associated peptide proteomics — identifies presented peptides to assess immunogenicity risk.
MRI	Magnetic resonance imaging
MRM / PRM	Multiple- and parallel-reaction monitoring — targeted, quantitative mass-spectrometry readouts.
MRS	Magnetic resonance spectroscopy
MST	Microscale thermophoresis — binding affinity from the motion of molecules in a temperature gradient.
NGS	Next-generation sequencing.
NMR	Nuclear magnetic resonance spectroscopy.
OCT	Optical coherence tomography
PAMPA	Parallel artificial membrane permeability assay — cell-free passive-permeability screen.
PBMC	Peripheral blood mononuclear cells.
PET	Positron emission tomography
PTM	Post-translational modification.
RNA-seq	RNA sequencing; scRNA-seq denotes the single-cell form.
RNAi	RNA interference — sequence-specific knockdown, used for genetic screens.
SEC-HPLC	Size-exclusion HPLC — aggregate and fragment quantification for protein therapeutics.
SMRT / HiFi	Single-molecule real-time long-read sequencing and its high-accuracy consensus reads.
SPECT	Single-photon emission computed tomography
SPR	Surface plasmon resonance — label-free, real-time binding kinetics and affinity.

TR-FRET	Time-resolved fluorescence resonance energy transfer — proximity-based homogeneous assay format.
WGBS	Whole-genome bisulfite sequencing — base-resolution DNA methylation mapping.
XL-MS	Crosslinking mass spectrometry — maps residue-level contacts in protein complexes.

Index of assays

ADMET - CYP450 inhibition/induction 11

ADMET - efflux transporter assays 11

ADMET - metabolic stability 10

ADMET - permeability (PAMPA/Caco-2) 10

ADMET - plasma protein binding 11

ADMET - plasma/blood stability 11

ADMET - solubility 10

Aggregation propensity (SEC-HPLC) 11

AlphaScreen / AlphaLISA proximity assays 9

Antimicrobial susceptibility testing (culture/AST) 12

Apoptosis assays 9

ATAC-seq 5

ATPase / GTPase activity assays 8

Bisulfite sequencing / WGBS 5

Bottom-up (shotgun) proteomics 6

Bulk RNA-seq 5

Cardiac MRI 3

Cell viability / cytotoxicity assays 9

Cell-based functional / receptor assays 9

Charge variant analysis (IEX) 11

Chemical proteomics / ABPP 6

ChIP-seq 5

Cone-beam CT 4

CRISPR knock-out/knock-in gene editing 8

Crosslinking-MS (XL-MS) 6

CT - non-contrast and contrast-enhanced 3

CT angiography 4

CUT&RUN / CUT&Tag 5

Digital pathology / whole-slide imaging 5

Digital radiography (X-ray) 4

Directed evolution / library screening	12
Dose-response / EC50 / IC50 determination	10
Dual-energy X-ray absorptiometry (DEXA)	4
Echocardiography	4
Electrophysiology / patch clamp (hERG, ion channels)	10
Enzyme inhibition assays (IC50/Ki)	8
Epitope binning, FcRn binding	11
Fc effector function assays (ADCC/CDC/ADCP)	11
Flow cytometry and immunophenotyping	9
Fluoroscopy and interventional angiography	4
Fluxomics	7
Forced degradation / stress stability studies	11
Fragment-based screening (NMR)	8
Functional agonist/antagonist assays	10
GC-MS based metabolomics	7
Genome-wide / boutique CRISPR & RNAi screens	8
Genotoxicity screening (Ames test)	10
HDX-MS	6
Hi-C / chromosome conformation capture	5
High-content imaging	3
High-throughput biochemical screening (HTS)	8
Histopathology and tissue staining	3
Hydrophobicity (HIC-HPLC)	11
In vitro immunogenicity (T-cell assays)	12
In vivo PK/PD, xenograft efficacy studies	10
ITC (isothermal titration calorimetry)	9
Kinase activity assays	8
Knockout/knock-in phenotypic characterisation	8
LC-MS based metabolomics (untargeted/targeted)	7
Long-read / full-length isoform sequencing	5
Mammography and digital breast tomosynthesis	4
Medicinal/peptide chemistry synthesis	12

Metabolic pathway engineering / product-titer assays 12
Metaproteomics 6
MRI - diffusion-weighted and DTI 3
MRI - functional (BOLD) 3
MRI - spectroscopy (MRS) 3
MRI - structural 3
NMR-based metabolomics 7
Off-target selectivity panels 10
Organism identification (MALDI-TOF) 12
PET/CT 4
PET/MRI 4
Phenotypic cell-based screening 9
Polyreactivity (ELISA) 11
Protease activity assays 8
Protein engineering stability assays 12
Protein-protein interaction mapping 9
Proximity labeling (BioID/APEX) 6
PTM-focused proteomics (phospho/glyco/ubiquitin/acetyl) 6
Quantitative bioanalysis 10
Recombinant protein expression (E. coli / CHO) 12
Reporter gene assays 9
Retinal imaging - fundus and OCT 4
Single-cell / single-nucleus RNA-seq 5
Single-cell proteomics 6
Small RNA-seq (miRNA) 5
Spatial proteomics (imaging MS) 3
Spatial transcriptomics 3
SPECT and SPECT/CT 4
SPR / BLI / MST (binding kinetics) 9
Targeted proteomics (MRM/SRM/PRM) 6
Targeted qPCR resistance-gene panels 12
Targeted RNA panels 5

Thermal shift / DSF for hit-binding confirmation 9
Thermal stability (Tm) - proteins/antibodies 11
Top-down / middle-down proteomics 6
TR-FRET assays 9
Tumour and tissue pharmacodynamics 10
Ultrasound - B-mode and Doppler 4
Ultrasound elastography 4
Viscosity assessment 11